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Describe the role of **two** named enzymes in the process of semi-conservative replication of DNA.

[3 marks]

[Extra space]

Question 9 continues on the next page

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09.2

Scientists investigated the function of a eukaryotic cell protein called cyclin A. This protein is thought to be involved with the binding of one of the enzymes required at the start of DNA replication.

The scientists treated cultures of cells in the following ways.

C – Control cells, untreated

D – Added antibody that binds specifically to cyclin A

E – Added RNA that prevents translation of cyclin A

F – Added RNA that prevents translation of cyclin A **and** added cyclin A protein

They then determined the percentage of cells in each culture in which DNA was replicating.

Their results are shown in **Table 5**.

Table 5

Cell treatment	Percentage of cells where DNA was replicating
C Control	91
D Antibody that binds specifically to cyclin A	11
E RNA that prevents translation of cyclin A	10
F RNA that prevents translation of cyclin A and added cyclin A protein	92

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[3 marks]

[illegible][illegible]

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